

Vol 4 Chapter 12 — The Observational Reanalysis Program (SP-27)

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Chapter 12 — The Observational Reanalysis Program (SP-27)

This is the closing chapter of Volume 4. The substrate's cosmological derivations across Chapters 1–11 produce many testable observables. SP-27 is the framework's structured program for systematic comparison of substrate predictions against observational data.

12.1 The SP-27 program

SP-27 (Casey directive May 16, 2026) is the Observational Reanalysis Program — a multi-year effort to reanalyze existing cosmological survey data (Planck, DES, eBOSS, etc.) against BST-substrate primary predictions, identifying where the substrate framework's predictions agree with observation, where they diverge, and where the divergences point to substrate-side refinement opportunities.

The current program scope (~30% complete) covers:

- CMB temperature + polarization spectra
- Supernovae Hubble-diagram analysis
- BAO scale measurements
- Cluster cosmology
- Weak-lensing surveys

12.2 Methodology

Per chapter analysis runs through:

1. Identify the substrate-derived prediction for the observable
2. Compare to observed value with reported uncertainties
3. Compute deviation in BST primary tier (D/I/C/S)
4. Flag for honest-scope refinement or D-tier ratification
5. Cross-reference to the catalog and the K-audit chain

12.3 Falsifier table

The chapter's payload is the per-observable falsifier table — what BST predicts, what would refute the prediction, current experimental status. Sample entries:

- H_0 : BST predicts $\sim 67\text{--}68$ km/s/Mpc (Chapter 5, four routes). Refuted if SH0ES distance ladder systematic is resolved and gives > 70 km/s/Mpc.
- n_s : BST predicts $1 - 5/137 = 0.9635$. Refuted if Planck-precision measurement diverges by $> 1\%$.
- Λ matches to 0.076 dex. Refuted by precision improvement showing > 0.5 dex deviation.
- DM/baryon ratio: BST predicts 16/3. Refuted by precision improvement showing $> 2\%$ deviation.
- r (tensor-to-scalar): BST predicts ≈ 0 . Refuted by detection of $r > 0.01$.

12.4 What comes next

Volume 5 — Quantum Mechanics from D_IV⁵ — develops the substrate-derivation of quantum mechanics' foundations, building on the Born=Bergman result of Volume 1 Chapter 7 and treating the measurement problem, Bell inequalities, decoherence, and POVMs at substrate-mechanical depth.

The reader who has worked through Volume 4 has now seen the substrate's cosmological derivations end-to-end. The cosmological constant is resolved at 99.94%; the Hubble tension has a substrate-side resolution; CMB structure matches; BBN matches; the dark sector emerges from substrate Shannon channels. The substrate produces cosmology from the same five integers that produce particle physics.

Where to look this up: SP-27 program: Casey directive May 16, 2026; multi-year ongoing. For cosmological observation reviews: Planck 2018 results; DES Y3 release.